Ver	Eff_Date	ECN No.	Author	Change Description
0.1	03-05-03	T0305011	S. W. Tsao	Original
0.2	04-18-03	T0314011	S. W. Tsao	1.update the title to 2P6M 2.add rule M5,M6 3.add rule via4,via5 4.update the FTV spec to(7.5,9) 5.update the resist of cont-top metal to (55,500)
0.3	03-26-04	E080200410023	S. W. Tsao	Update the parameter name.
0.4	04-22-04	E080200415016	S. W. Tsao	Update the spec of Ir to (20,40) asVg=2.5V
0.5	05-05-04	E080200418002	S. W. Tsao	Update the HV device spec measured at Vg=1.8V
1.0	06-07-04	E080200421023	S. W. Tsao	Update the version to 1.0
1.1	07-09-04	E080200426033	S. W. Tsao	Update the measurement method
1.2	08-20-04	E080200433024	S. W. Tsao	Update HV Idsat measurement method to Vg=Vd=2.5V

Revisor : S. W. Tsao (NVP)

Revising Dept. Manager : Y. T. Lin

Approvals :

ATM	: S. J. Han	ENVM	: K. Y. Lee
SPICE	: C. K. Linb	DSD	: M. I. Chang
APE1	: Y. J. Chu	MM_RF	: C. S. Hsua
MM_RF	: C. F. Chene	PDSE	: T. H. Hsiehb

★ Signature on file in DC ★

Title

TSMC 0.18 UM CMOS EMBEDDED MEMORY FLASH LOGIC BASED GENERAL PURPOSE 2P6M SALICIDE AL_FSG 1.8&3.3V PCM SPEC

Document No. : T-018-CE-PC-001

Contents	:2	Review (Date & Sig.) :
Attach.	:0	
Total	:2	
File Format	:W	

★ PLEASE RETURN OLD VERSION SPEC TO DC ★

The information contained herein is the exclusive property of TSMC and shall not be distributed, copied, reproduced, or disclosed in whole or in part without prior written permission of TSMC.

TSMC 0.18 UM CMOS EMBEDDED MEMORY FLASH LOGIC BASED GENERAL PURPOSE 2P6M SALICIDE AL_FSG 1.8&3.3V PCM SPEC

testline:TL0320,TL0561

Parameter	Device Size	Spec Low	Spec High	Unit	Measurement Method
Vt_N4 EXT	W/L 10/0.18	.32	.52	V	const ant current method
Isat_N4	W/L 10/0.18	5.1	6.9	mA	Vg=Vd=1.8V, Vb=Vs=GND
BV_N4	W/L 10/0.18	3.6	N/A	V	Vg=Vb=Vs=GND Id =1uA
VTFpo_N+	W/L 10/0.28	3.6	N/A	V	Vd=2V Vb=Vs=GND, Id = 0.1 uA Sense Vg
Vt_P4 EXT	W/L 10/0.18	-0.6	-0.4	V	constant current method
Isat_P4	W/L 10/0.18	-2.99	-2.21	mA	Vg=Vd=-1.8V, Vb=Vs=GND
BV_P4	W/L 10/0.18	N/A	-3.6	V	Vg=Vb=Vs=GND Id=-1uA
VTFpo_P+	W/L 10/0.28	N/A	-3.6	V	Vd=-2V Vb=Vs=GND Id=-0.1 uA Sense Vg
VT_N43 EXT	W/L 10/0.35	0.62	0.82	V	constant current method
Isat_N43	W/L 10/0.35	5.1	6.9	mA	Vg=Vd=3.3V, Vb=Vs=GND
BV_N43	W/L 10/0.35	6.0	N/A	V	Vg=Vb=Vs=GND Id =1uA
VT_P43 EXT	W/L 10/0.3	-0.78	-0.58	V	constant current method
Isat_P43	W/L 10/0.3	-3.45	-2.55	mA	Vg=Vd=-3.3V, Vb=Vs=GND
BV_P43	W/L 10/0.3	N/A	-6.0	V	Vg=Vb=Vs=GND Id=-1uA
Rs_N+	W/L 22/100	1	15	Ohm/Sq	Force V=1.8V Sense I
Rs_P+	W/L 22/100	1	15	Ohm/Sq	Force V=1.8V Sense I
Rs_N+PO	W/L 18/100	1	20	Ohm/Sq	Force V=1.8V Sense I
Rs_P+PO	W/L 18/100	1	20	Ohm/Sq	Force V=1.8V Sense I
Rs_NW(STI)	W/L 20/100	0.6	1.4	Kohm/Sq	Force V=1.8V Sense I
Rc_N+	CO =0.22x 0.22	1	30	Ohm/Sec	Force V=1.8V Sense I
Rc_P+	CO =0.22x 0.22	1	30	Ohm/Sec	Force V=1.8V Sense I
Rc_N+PO	CO =0.22x 0.22	1	30	Ohm/Sec	Force V=1.8V Sense I
Rc_P+PO	CO =0.22x 0.22	1	30	Ohm/Sec	Force V=1.8V Sense I
Rc_Via1	VIA =0.26x 0.26	1	20	Ohm/Sec	Force V=1.8V Sense I
Rc_Via2	VIA =0.26x 0.26	1	20	Ohm/Sec	Force V=1.8V Sense I
Rc_Via3	VIA=0.26x0.26	1	20	Ohm/Sec	Force V=1.8V Sense I
Rc_Via4	VIA=0.26x0.26	1	20	Ohm/Sec	Force V=1.8V Sense I
Rc_Via5(top via)	VIA =0.36x 0.36	1	13	Ohm/Sec	Force V=1.8V Sense I
CONTI_PO	W/L=0.18/1756	10	150	Kohm	Force V=1.8V Sense I
CONTI_M1	W/L=0.23/1756	100	1000	Ohm	Force V=1.8V Sense I
CONTI_M2	W/L=0.28/1756	100	900	Ohm	Force V=1.8V Sense I
CONTI_M3	W/L=0.28/1756	100	900	Ohm	Force V=1.8V Sense I
CONTI_M4	W/L=0.28/1756	100	900	Ohm	Force V=1.8V Sense I
CONTI_M5	W/L=0.28/1756	100	900	Ohm	Force V=1.8V Sense I
CONTI_M6(top metal)	W/L=0.44/1756	55	500	Ohm	Force V=1.8V Sense I
SPA_PO/PO	SPACE=0.25	7	N/A	V	Force I=1uA Sense V

The information contained herein is the exclusive property of TSMC and shall not be distributed, copied, reproduced, or disclosed in whole or in part without prior written permission of TSMC.

SPA_M1/M1	SPACE=0.23	7	N/A	V	Force I=1uA Sense V
SPA_M2/M2	SPACE=0.28	7	N/A	V	Force I=1uA Sense V
SPA_M3/M3	SPACE=0.28	7	N/A	V	Force I=1uA Sense V
SPA_M4/M4	SPACE=0.28	7	N/A	V	Force I=1uA Sense V
SPA_M5/M5	SPACE=0.28	7	N/A	V	Force I=1uA Sense V
SPA_M6/M6(top metal)	SPACE=0.46	7	N/A	V	Force I=1uA Sense V
VBD_GO/PW	A=11200um^2	N/A	-4.6	V	Pulsed sweep V sense I
VBD_GO/NW	A=11200um^2	4.6	N/A	V	Pulsed sweep V sense I
VBD_GO/PW(3.3V)	A=11200um^2	N/A	-8	V	Pulsed sweep V sense I
VBD_GO/NW(3.3V)	A=11200um^2	8	N/A	V	Pulsed sweep V sense I
VBD_GO/Psub	A=93800um^2	-40	-15	V	Pulsed sweep V sense I
VBD_GO/HVNW	A=93800um^2	15	40	V	Pulsed sweep V sense I
Vt_HN2 EXT	W/L 10/1.0	0.4	0.7	V	Linear Extrapolation (Vd=0.1 V,
					Vb=Vs=GND, Sweep Vg till GM Max)
Isat_HN2	W/L 10/1.0	1.06	1.5	mA	Vg=Vd=2.5V, Vb=Vs=GND
BV_HN2	W/L 10/1.0	12	20	V	Vg=Vb=Vs=GND Id =0.1uA
VTFpo_HN1	W/L 10/1.0	14	20	V	Vd=10V Vb=Vs=GND, Id = 100 nA Sense Vg
BVFm1_HN1	W/L 10/1.0	13	20	V	Vg=Vb=Vs=GND, Id = 100 nA Sense Vd
Vt_HP2 EXT	W/L 10/0.9	-0.84	-0.56	V	Linear Extrapolation (Vd=-0.1 V,
					Vb=Vs=GND, Sweep Vg till GM Max)
Isat_HP2	W/L 10/0.9	-0.53	-0.31	mA	Vg=Vd=-2.5V, Vb=Vs=GND
BV_HP2	W/L 10/0.9	-20	-12	V	Vg=Vb=Vs=GND Id=-100nA
					(with Id=-1uA stress@Vg=Vb=Vs=GND for 100ms)
BVFm1_HP1	W/L 10/1.0	-30	-13	V	Vg=Vb=Vs=GND, Id = -100 nA Sense Vd
Vt_ZN2 EXT	W/L 10/1.5	0.18	0.32	V	Linear Extrapolation (Vd=0.1 V,
					Vb=Vs=GND, Sweep Vg till GM Max)
Isat_ZN2	W/L 10/1.5	1.09	1.49	mA	Vg=Vd=2.5V, Vb=Vs=GND
Ir1_T	P1/P2 0.26/0.42	20	40	uA	Vg=2.5V, Vd=0.6V, Vb=Vs=GND,
					Erase: Vee=11V, 10ms
Ir1_B	P1/P2 0.26/0.42	20	40	uA	Vg=2.5V, Vd=0.6V, Vb=Vs=GND,
					Erase: Vee=11V, 10ms
Ir0_T	P1/P2 0.26/0.42	0	3	uA	Vg=2.5V, Vd=0.6V, Vb=Vs=GND,
					Program: Idp=1.5uA, Vsp=9.5V, 10us
Ir0_B	P1/P2 0.26/0.42	0	3	uA	Vg=2.5V, Vd=0.6V, Vb=Vs=GND,
					Program: Idp=1.5uA, Vsp=9.5V, 10us
BV_SP	P1/P2 0.26/0.42	9	20	V	Vg=Vb=GND, Vd=VdP, Is =0.1uA
FTV	512 cells	7.5	9	V	Iforce=10nA for 0.2sec with V compliance=4V,
					Then Iforce=10nA for 0.1sec, measure tunnel voltage
RTV	512 cells	12	20	V	Iforce@-10nA for 0.1sec, then measure tunnel voltage
TUR	512 cells	0	.6	V	[FTV(5)-FTV(0.3)]/log(50)
Read_2	1024 cells	0	100	nA/kbit	BL0=1.8V;BL1=0V;Vwl=Vss=Vb=0V;measure I(BL0)
Rs_SL_RPO	cell array	60	120	Ohm/Cell	Force V=1V Sense I

Remark:
1. for 1P5M process, please skip M5,and Via4 spec., and treat M6 as top metal.
2. for 1P4M process, please skip M5,M4,Via4,and Via3 spec., and treat M6 as top metal.
3. for 1P3M process, please skip M5,M4,M3,Via4,Via3,Via2 and Via1 spec., and treat M6 as top metal

The information contained herein is the exclusive property of TSMC and shall not be distributed, copied, reproduced, or disclosed in whole or in part without prior written permission of TSMC.